



Earth without ice

What would the Earth look like if all the ice in the polar caps melted? This video shows the rise in water levels. <https://digit.in/may19-55>



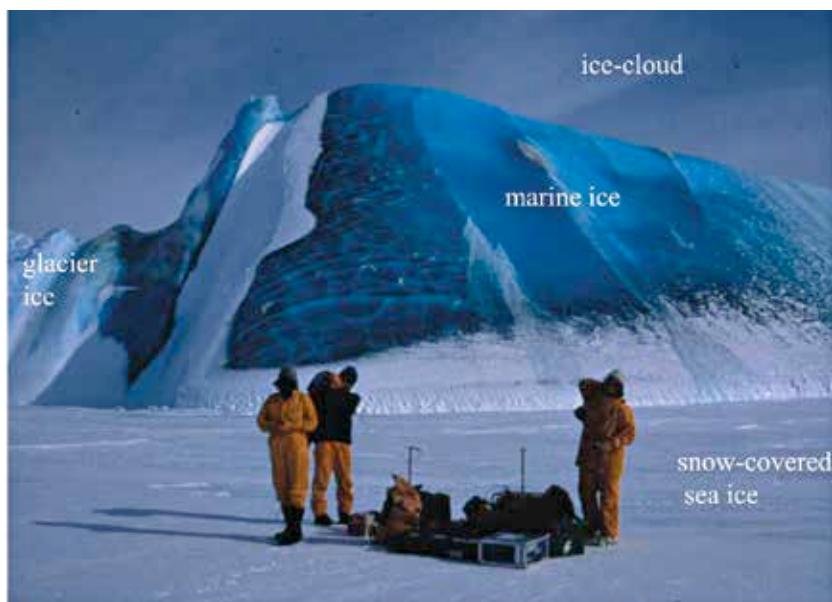
Life before Earth formation

Life on Earth might have evolved in the protoplanetary disk, before the Earth was fully formed. <https://digit.in/may19-56>

has been published in the Journal of Geophysical Research and is led by researchers from the University of Washington, supported by scientists from Bowdoin College and the Australian Antarctic Division.

Stephen Warren, the lead author of the study started investigating the phenomenon back in 1988. A core sample was drilled from an emerald iceberg off the Amery Shelf. The colouration was not the most striking aspect of the sample. Warren said, “when we climbed up on that iceberg, the most amazing thing was actually not the color, but rather the clarity. This ice had no bubbles. It was obvious that it was not ordinary glacier ice.” At that time, the researchers suspected that the colouration was because of some kind of impurity in the waters of the ocean, below the Amery shelf.

A big piece of the puzzle fell into place when an oceanographer from the University of Tasmania, Dr Laura Herraiz-Borreguero discovered surprisingly high amounts of iron in samples retrieved from the Amery Ice Shelf. The ice near the bottom of the marine ice sample had nearly five hundred times more iron than the glacier ice on top. The finding prompted the University of Washington researcher Stephen Warren and his colleagues to propose that the iron content was what was giving the icebergs the green colouration. Iron oxides are rust coloured, and are yellow, red, orange or brown in colour. When combined



The different kinds of ice in a glacier

Credit: Steve Warren/University of Washington

with the blue of the marine ice, they cause the green colouration. Dr Mark Curran, co-author of the study says, “I first saw one of these intriguing icebergs from the deck of the Aurora Australis in Prydz Bay in 1992, and no one could tell me what caused the colour. It’s a question that has intrigued many seagoing Antarctic scientists and explorers over the years, and so it was a real honour to play a part in this fascinating piece of research.”

NUTRITION FOR MARINE LIFE

The new research also investigates the process by which the marine ice acquires the iron content. When the glacial ice flows over the bedrock, it grinds the ground below, forming a powder known as glacial flour, or rock flour. When the ice sheets then flow over the water, this glacial flour gets trapped in the newly formed marine ice below the glacial ice. While the green icebergs were so far considered just scientific curiosity, the new study shows that they may play a vital role in the ecosystem. As the icebergs float out into the sea and melt down, the iron gets released deep into the oceans, providing essential nutrients to marine life. The marine life that benefits from this source of nutrition potentially include phyto-

plankton blooms, microscopic algae that are at the bottom of the oceanic food chain. These blooms are massive and can stretch across oceans. Warren explains the process as, “It’s like taking a package to the post office. The iceberg can deliver this iron out into the ocean far away, and then melt and deliver it to the phytoplankton that can use it as a nutrient. We always thought green icebergs were just an exotic curiosity, but now we think they may actually be important.”

Phytoplankton blooms support a lot of marine life, and iron is not that easily available in the oceans. The green icebergs may be playing an important role in the marine ecosystem, by providing the nutrition to the phytoplankton, which in turn support nearly all marine life we know of!

At this point, the proposed processes that lead to the green icebergs are just a theory. The team of researchers plan to obtain samples of icebergs of different colours, and observe them for their light reflecting properties, as well as iron content. This further study will allow the team to confirm or deny their suspicions. If the theory proves to be true, then the green icebergs may turn out to be a vital component of the ecosystem, and not just another pretty looking natural anomaly. 



Credit: Steve Nicol

A jade iceberg sculpted by the wind and waves